

Windows Server Deployment

Active Directory Forest, DNS/DHCP and Group Policy Baselines

Marouane Aabirrouche — IT Infrastructure specialist — Réseaux & Systèmes
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Overview

This project covers the design and deployment of a complete Windows Server environment built around a full *Active Directory* forest with integrated *DNS* and *DHCP*, locked down by strict *Group Policy Objects* (GPOs) providing comprehensive user management and domain-wide security baselines. The goal was to reproduce, end to end, the core identity and configuration-management layer that a small-to-medium enterprise runs on, and to validate that every control actually applies where it should.

1 Objectives

- Deploy a full Active Directory forest from a clean Windows Server install.
- Integrate DNS and DHCP into the directory so addresses, names and domain membership are managed centrally.
- Enforce comprehensive user management through organizational units (OUs) and security groups.
- Apply strict GPOs forming a security baseline across the domain: password policy, account lockout, audit policy and hardening settings.
- Verify enforcement empirically (gpresult / RSOP) instead of assuming it.

2 Environment

The lab runs virtualized on the home server estate alongside the Docker/Linux workloads described in the HomeLab report (PRJ-03). A dedicated Windows Server virtual machine acts as domain controller; client machines join the domain to exercise the policies.

Component	Role
Windows Server VM	Domain controller: AD DS, DNS, DHCP
Domain	Dedicated lab forest, single domain tree
Clients	Domain-joined workstations for policy validation
Management	Server Manager, Group Policy Management Console (GPMC), RSOP

3 Implementation

3.1 Forest and domain design

The deployment starts from a clean installation: static addressing for the server, then promotion to domain controller, which installs the Active Directory Domain Services role and the integrated DNS zone. A single-domain forest keeps the trust model simple while leaving room for future child domains. Default containers were left untouched and all administration happens against a deliberate OU hierarchy, so inheritance is predictable.

3.2 Organizational units and user management

OUs separate workstations, servers, users and service accounts. Security groups map to functional roles rather than individuals, and permissions are granted to groups only — a pattern that keeps access reviews trivial. User accounts are created centrally, with group membership driven by OU placement and role mapping.

3.3 Integrated DNS and DHCP

DNS lives inside the directory: the AD-integrated zone replicates with the database itself, domain members resolve internal names reliably, and the server forwards external queries upstream. DHCP is authorized in AD and serves the lab scope with reservations where stable addressing matters; clients receive the domain controller as their DNS server so domain joins and name resolution “just work”.

3.4 Group Policy baselines

Policy is applied in layers — domain-level baseline first, then OU-specific refinements:

- **Password and lockout policy:** length and complexity requirements, history, lockout thresholds and durations.
- **Account hardening:** restriction of local administrator use, controlled folder and service permissions.
- **Audit policy:** logon/account events and object access auditing feeding the event logs.
- **User experience restrictions:** drive mappings, desktop and start menu lockdowns where appropriate, demonstrating per-audience control.

4 Validation

Every GPO was validated with `gpresult` / Resultant Set of Policy on target machines: password changes rejected when below baseline, lockout firing after repeated failures, mapped drives appearing only for the intended groups, and audit entries landing in the Security log. DNS resolution was tested for internal records and forwarded lookups; DHCP leases, renewals and reservations were confirmed from the client side.

5 Outcomes and lessons learned

- Identity, name resolution and address assignment integrate cleanly when DNS/DHCP live with the directory — split-brain issues disappear.
- GPO layering rewards discipline: baseline at the domain, exceptions at the OU, nothing enforced ad hoc on individual machines.
- Testing policy application (RSOP/`gpresult`) after each change catches scope mistakes immediately — the difference between “applied” and “should apply”.